LEARNER GUIDE

Simplifying algebraic fractions

Algebraic Expressions and Identities · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

15/21 simplifies because 15 = 3 × 5 and 21 = 3 × 7: the common **factor** 3 cancels, since 3/3 = 1. Algebraic fractions work the same way. First **factorise** the numerator and the denominator; then cancel any factor they share — a number, a letter or a whole bracket. Only factors cancel, never terms: $(x + 3)/3$ is not $x + 1$, because the 3 is added, not multiplied. A value of x that makes the original denominator zero must be excluded.

What you are learning to notice

- Write the numerator and the denominator as products before cancelling anything.
- A whole bracket can cancel: $(x + 2)(x + 3)/(x + 2) = x + 3$.
- A term inside a bracket cannot: $(x + 5)/(x + 7)$ does not simplify.
- Test a shortcut with a number: at $x = 3$, $(x + 3)/3 = 2$ but $x + 1 = 4$, so that cancelling is wrong.
- Note excluded values: $(2x + 6)/(x^2 - 9) = 2/(x - 3)$, for $x \neq \pm 3$.

Moving through it, one step at a time

Factorise the numerator and the denominator fully. Cancel each factor that appears in both — one from the numerator with one from the denominator. Multiply what is left. State any values of x that would make the original denominator zero. Check by putting in a number.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Factorise, then cancel

Simplify $(2x + 6)/(x^2 - 9)$.

Factor tiles

2	~~$(x + 3)$~~
~~$(x + 3)$~~	$(x - 3)$

1. Factorise the numerator.	$2x + 6 = 2(x + 3)$
2. Factorise the denominator: a difference of two squares.	$x^2 - 9 = (x + 3)(x - 3)$
3. Cancel the common factor $(x + 3)$.	$(x + 3)/(x + 3) = 1$
4. What is left.	$2/(x - 3)$

$(2x + 6)/(x^2 - 9) = 2/(x - 3)$, for $x \neq \pm 3$.

Example 2 — Two quadratics

Simplify $(x^2 - 4)/(x^2 + 4x + 4)$.

Factor tiles

~~$(x + 2)$~~	$(x - 2)$
~~$(x + 2)$~~	$(x + 2)$

1. Factorise both.	$(x + 2)(x - 2)$ over $(x + 2)(x + 2)$
2. One factor of $(x + 2)$ cancels with one.	the other $(x + 2)$ stays
3. What is left.	$(x - 2)/(x + 2)$
4. Check with $x = 1$: $(1 - 4)/(1 + 4 + 4) = -3/9$, and $(1 - 2)/(1 + 2) = -1/3$.	✓

$(x^2 - 4)/(x^2 + 4x + 4) = (x - 2)/(x + 2)$, for $x \neq -2$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (simplifying number fractions, and factorising), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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